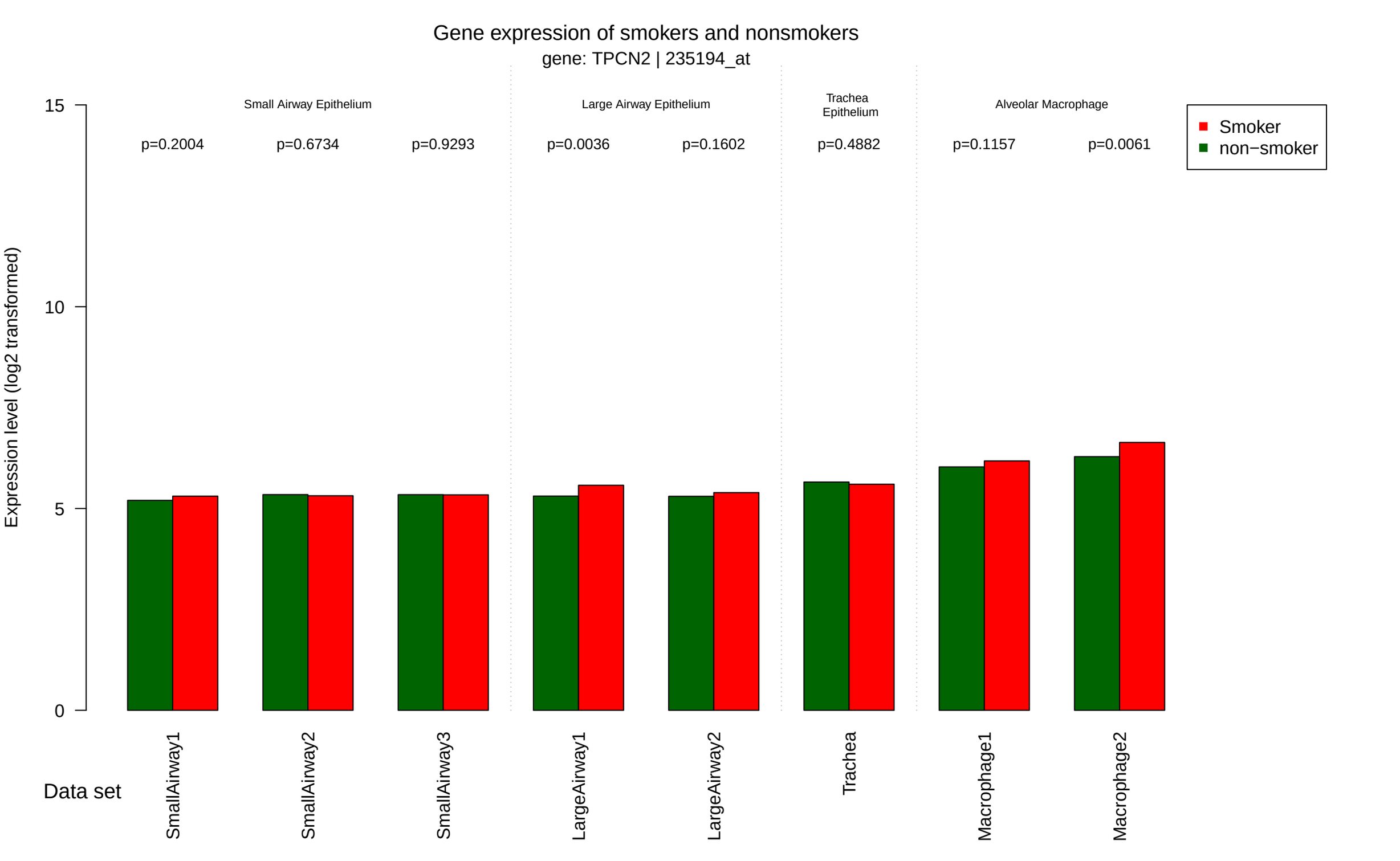
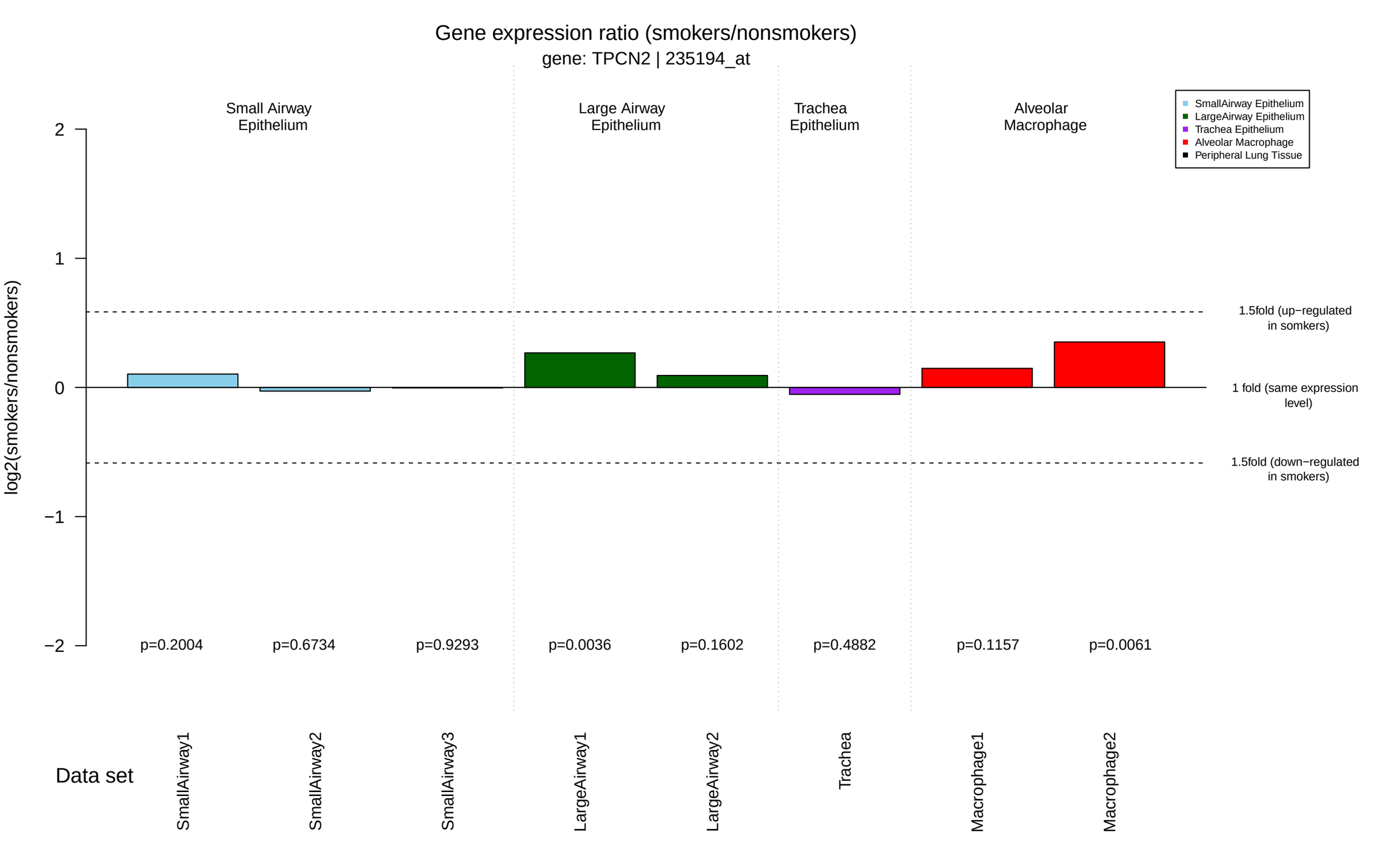
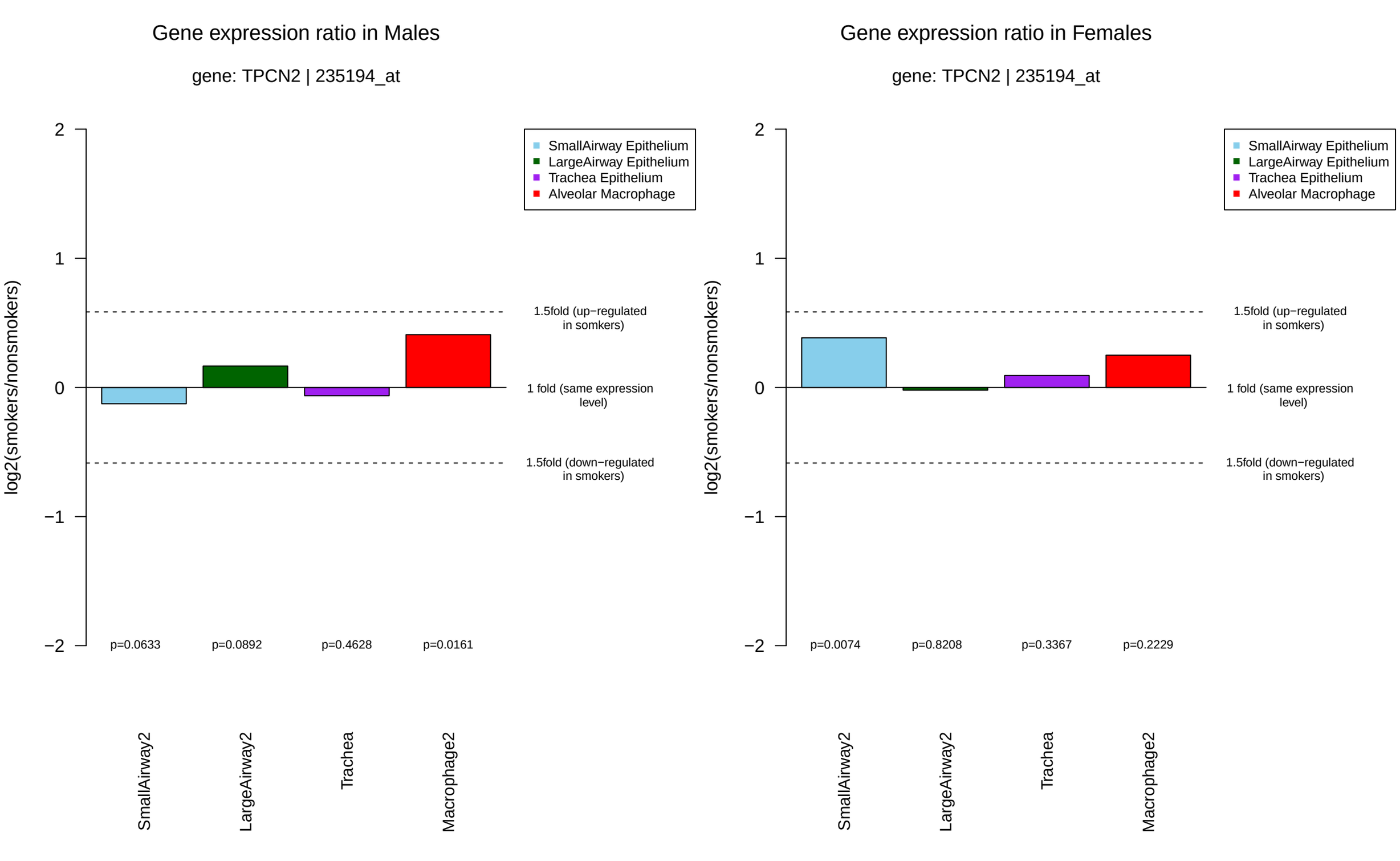
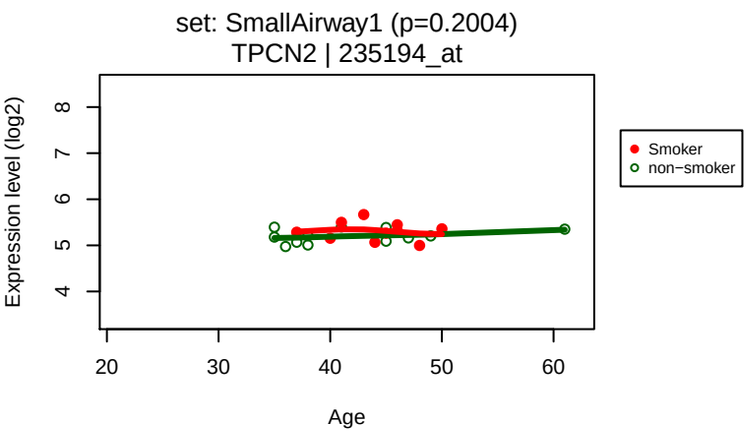
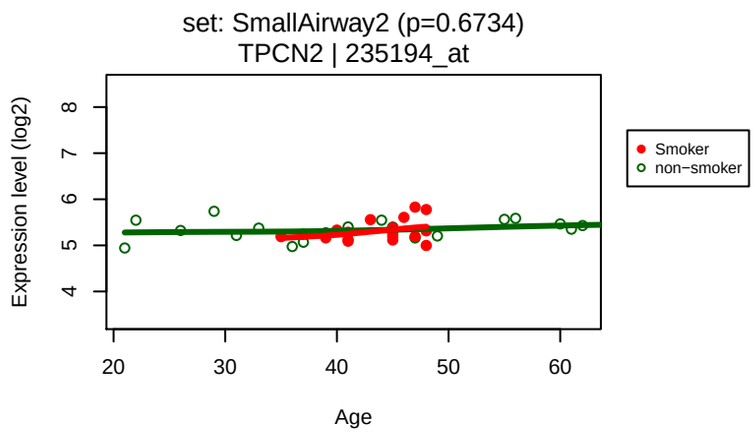
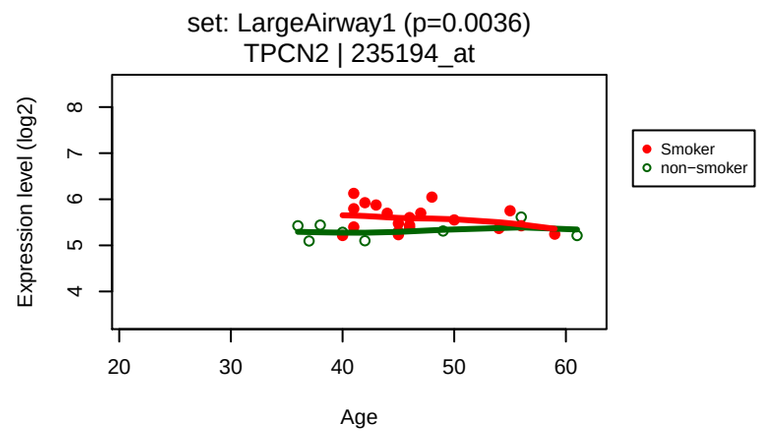

Gene expression of smokers and nonsmokers
gene: TPCN2 | 235194_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2004
p=0.6734
p=0.9293
p=0.0036
p=0.1602
p=0.4882
p=0.1157
p=0.0061
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

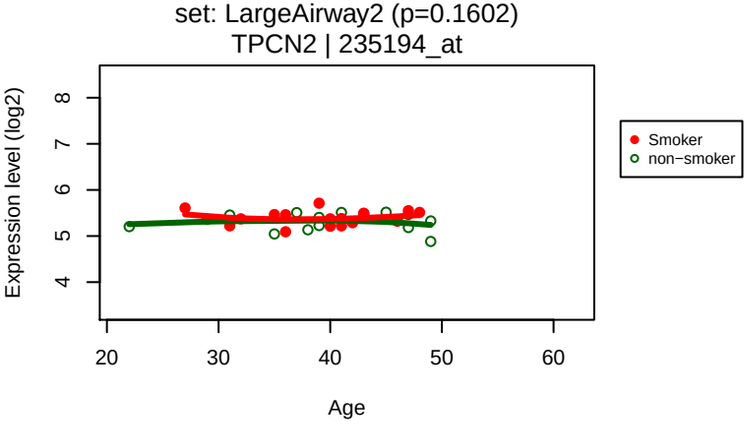

Gene expression ratio (smokers/nonsmokers)
gene: TPCN2 | 235194_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2004
p=0.6734
p=0.9293
p=0.0036
p=0.1602
p=0.4882
p=0.1157
p=0.0061
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

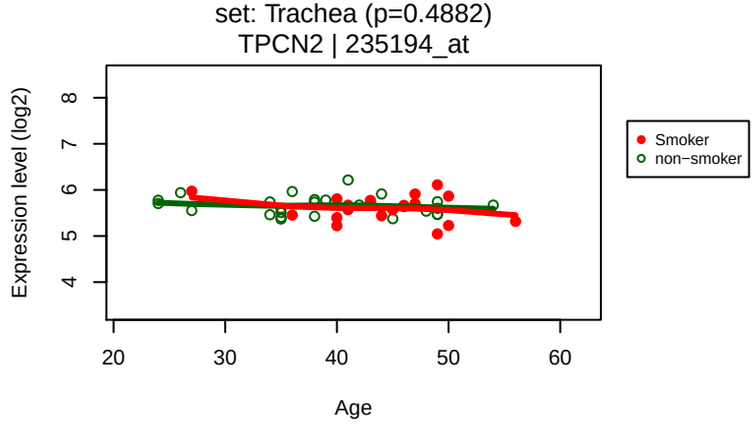

Gene expression ratio in Males
gene: TPCN2 | 235194_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0633
p=0.0892
p=0.4628
p=0.0161
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TPCN2 | 235194_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0074
p=0.8208
p=0.3367
p=0.2229
SmallAirway2
LargeAirway2
Trachea
Macrophage2

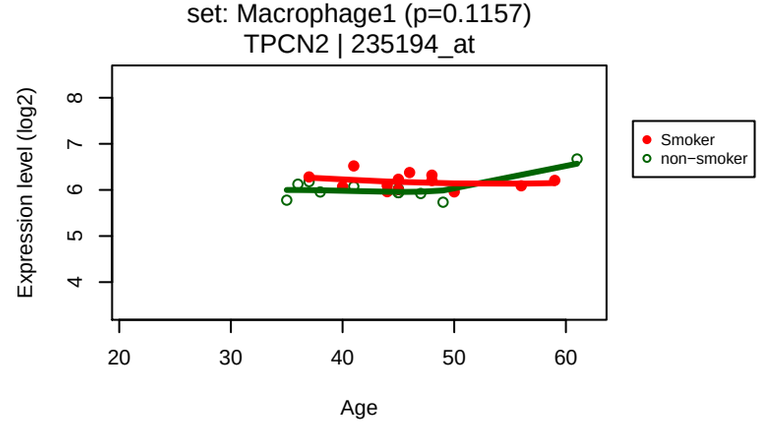

set: SmallAirway1 (p=0.2004)
TPCN2 | 235194_at
Expression level (log2)
Age
Smoker
non−smoker

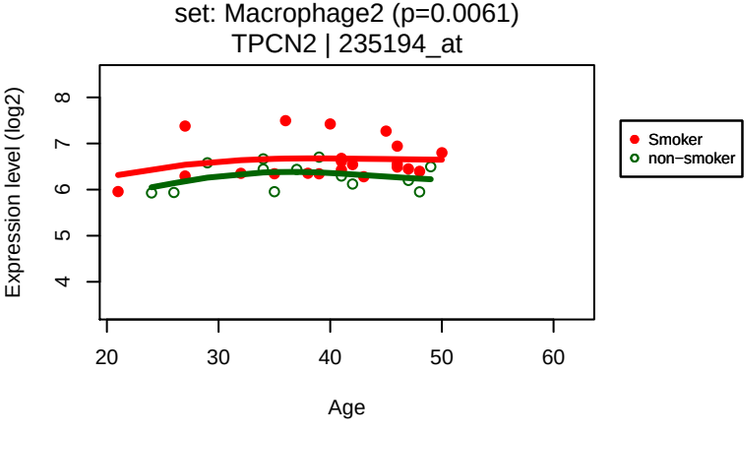

set: SmallAirway2 (p=0.6734)
TPCN2 | 235194_at
Expression level (log2)
Age
Smoker
non−smoker

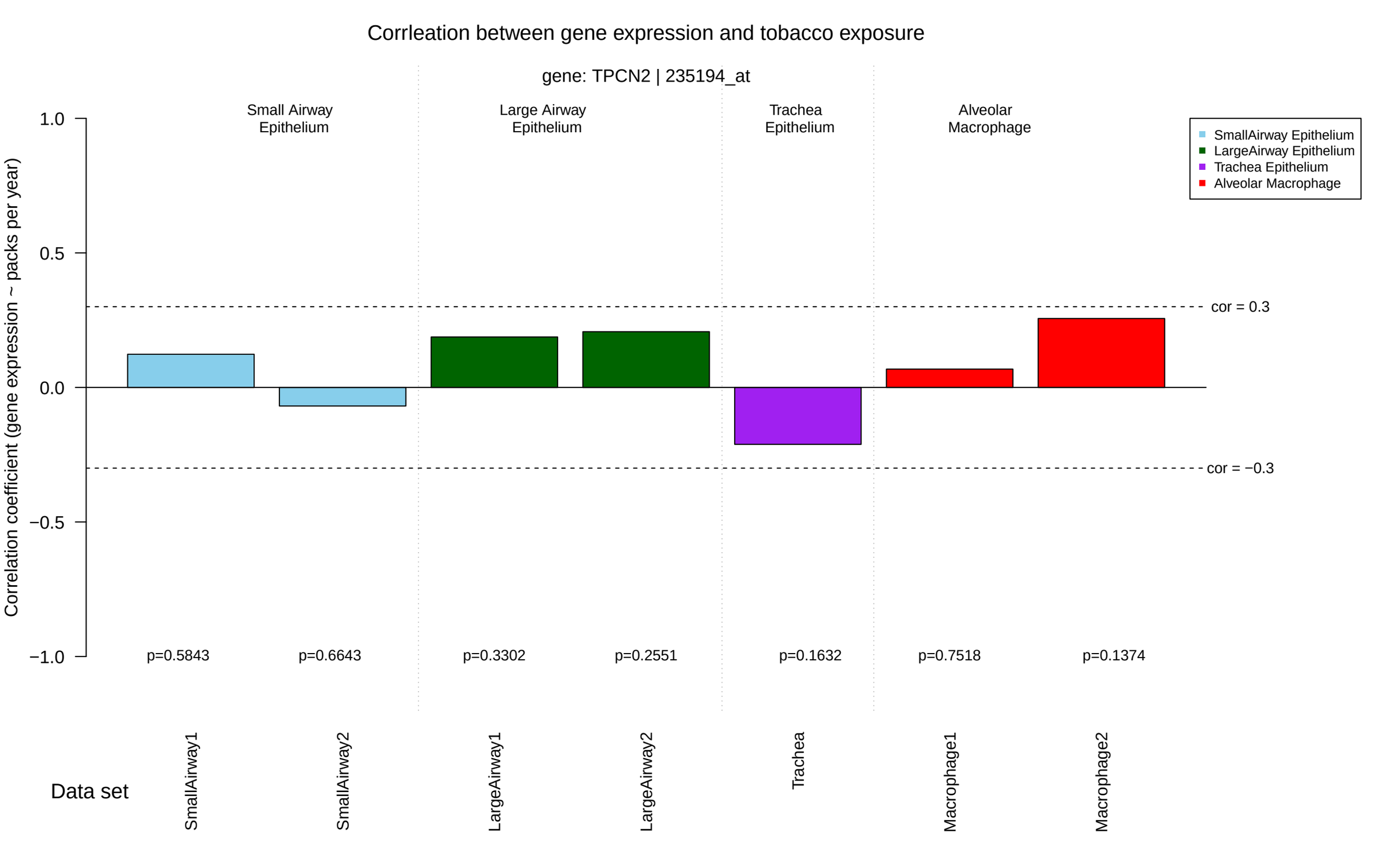

set: LargeAirway1 (p=0.0036)
TPCN2 | 235194_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1602)
TPCN2 | 235194_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4882)
TPCN2 | 235194_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1157)
TPCN2 | 235194_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0061)
TPCN2 | 235194_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TPCN2 | 235194_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5843
p=0.6643
p=0.3302
p=0.2551
p=0.1632
p=0.7518
p=0.1374
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2